AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 26 · WORKBOOK

Prompts Are Design Instructions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected prompt-design and testing cycle.

Clarify the club request, score an invented time, handle a room conflict and write the independent exhibit contract.

Keep one habit: Keep instructions, test inputs, outputs and criterion results together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Generate, Retrieve, or Calculate?

Fair for Whom?

Recommendations

W1 · Replace vague wishes with observable requirements

> Use the supplied C/P1 reference and explain each design choice.

Your evidence record

- Club-card reference: C says Monday, Room 7, bring paper; no time is supplied. P1 requires exactly four labelled lines in order Day, Time, Place, Bring, using C only, writing “not stated” for missing values and adding no other facts. P2 authorises the supplied records, preserves these requirements and writes “conflicting information” when equally authoritative records disagree without a stated priority. D describes the same event as C but gives Room 9; both agree on Monday and paper and neither supplies a time.

- The request says “Write something good about our club.” Name four decisions it leaves open. Then explain how source C, the Grade 6 **audience** and four labelled lines make the task easier to check.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Score format and evidence separately

> A passing layout does not excuse an invented time.

Your evidence record

- For original C, a card gives four correctly labelled lines but says Time: 3 pm. All other values match C. State which format and **evidence** checks pass or fail. Is the card acceptable overall? Explain.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Preserve a conflict without losing agreed facts

> Use P2 and the explicit equally authoritative source disagreement.

Your evidence record

- C gives Room 7 and D gives Room 9 for the same club event, with no replacement or priority stated. Both agree on Monday and paper, and neither gives a time.

  Under P2, write the four **output** values and explain why the room cannot be selected from confidence or wording alone.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a fairer small prompt comparison

> Retain all trials and use the same cases, criteria and recorded conditions.

Your evidence record

- A pupil tries P1 once and P2 five times, then compares the P1 response only with the best P2 response and declares P2 always reliable. Identify two problems and design a fairer small comparison.

  State a conclusion the **evidence** could support.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Specify and test a new exhibit card

> Keep one first attempt before hints or feedback.

Read and explain independently

Fictional record X says an exhibit is called Shadow Pairs and is in Hall 2; no opening time is given. Write a complete **prompt** for a Grade 6 card with exactly three labelled lines: Title, Location, Opening.

Limit facts to supplied records, define missing and conflicting values and prohibit added claims. Give two test **input**s and expected **output**s: original X, then X with an opening time of 10 am added.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Specify and test a new exhibit card

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Write a transparent instruction for any generative explanation in the sorter and forbid invented item facts.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Score a response by separate criteria for Specifying and testing **prompt**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Specifying and testing **prompt**s. Proposed explanation: “A tidy response is a correct response”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Write a request for a three-bullet explanation of evaporation for Grade 6 using only a supplied paragraph. Create three separately checkable acceptance criteria.

What should happen if the paragraph does not explain evaporation?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

The **prompt** says make it good; the desired **output** is a four-line rain poem using simple words.

Write a checkable **prompt**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · A changed case

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Improve this **prompt**: tell me about space. Specify a reader, a focused task and an **output** format suitable for a young learner.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 26 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Write a request for a three-bullet explanation of evaporation for Grade 6 using only a supplied paragraph. Create three separately checkable acceptance criteria.

What should happen if the paragraph does not explain evaporation?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Fictional record X says an exhibit is called Shadow Pairs and is in Hall 2; no opening time is given. Write a complete **prompt** for a Grade 6 card with exactly three labelled lines: Title, Location, Opening.

Limit facts to supplied records, define missing and conflicting values and prohibit added claims. Give two test **input**s and expected **output**s: original X, then X with an opening time of 10 am added.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

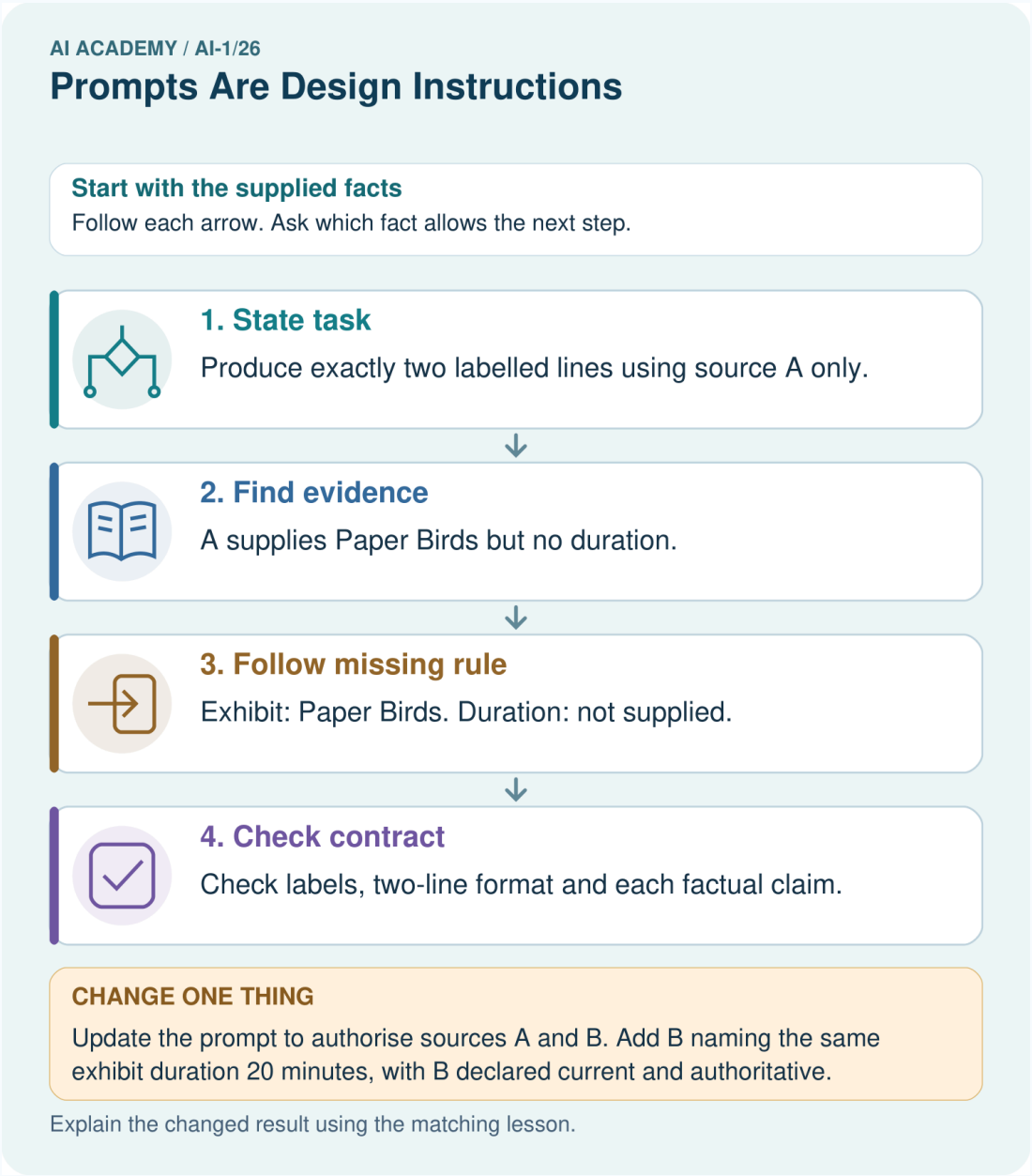
> Keep the explanation and the support you used.

Use this when helpful

A fictional club needs a short exhibit card, without guesses.

Source A: exhibit Paper Birds, location Shelf 2. No duration appears.

Required **output**: two labelled lines, Exhibit and Duration; missing values must say not supplied.



Follow the labelled picture with the matching explanation. Open the vector diagram

What must happen if B instead conflicts with another equally authoritative duration record?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Specify sorter **input**, **output** fields, **evidence** limits and missing/conflict behavior before improving its wording.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Specify sorter **input**, **output** fields, **evidence** limits and missing/conflict behavior before improving its wording.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional club needs a short exhibit card, without guesses.
- Which labelled step matches this detail? Check labels, two-line format and each factual claim.
- 1. State task
- 2. Follow missing rule
- 3. Find **evidence**
- 4. Check contract
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Check contract Check labels, two-line format and each factual claim. A useful **prompt** describes both the wanted answer and what to do when **evidence** is insufficient.

Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts. Check content support separately from whether the answer looks tidy.

Changed-case question

- Changed-case practice: Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Duration becomes 20 minutes.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Duration becomes 20 minutes. There is now an authorised source for the previously missing field.

Repair and save your work

Review Generate, Retrieve, or Calculate? (ai-1/25). Goal: Distinguish retrieving a record, calculating from **input**s and generating wording. Short practice: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?. Return to this week and explain what you changed.

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?.

Return to this week and explain what you changed.

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone’s preferences?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

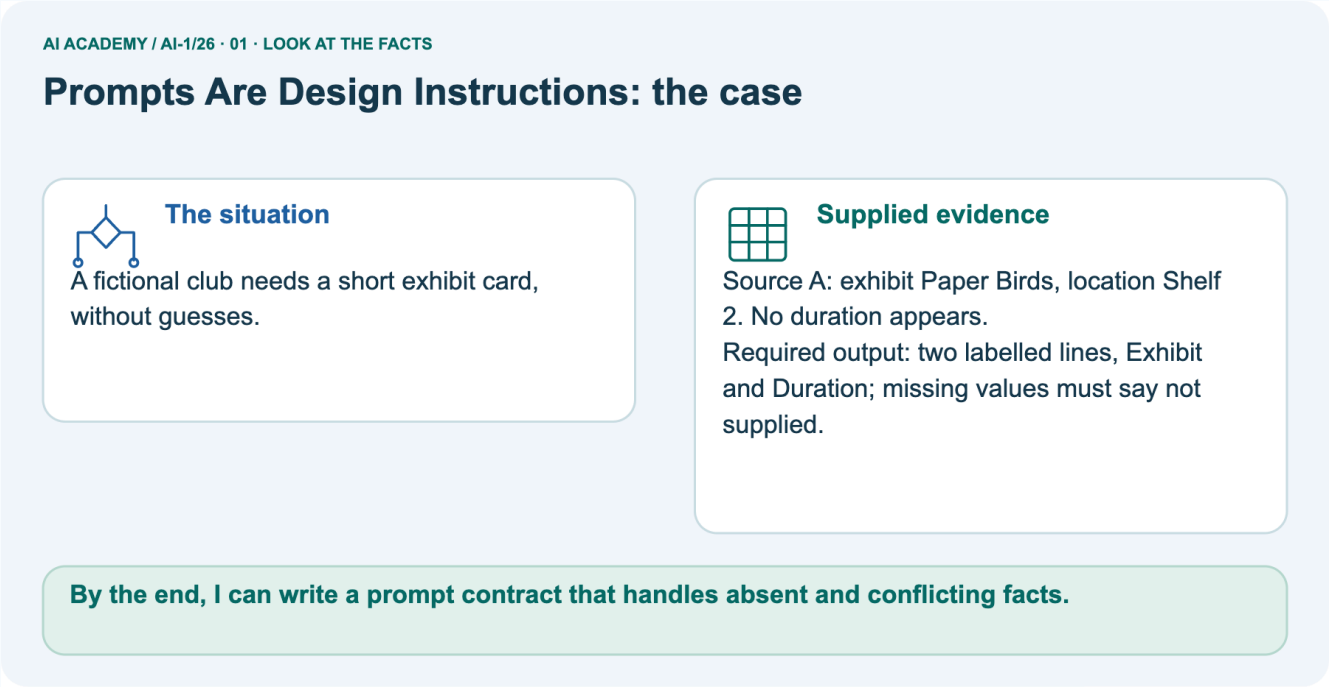
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Prompts Are Design Instructions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

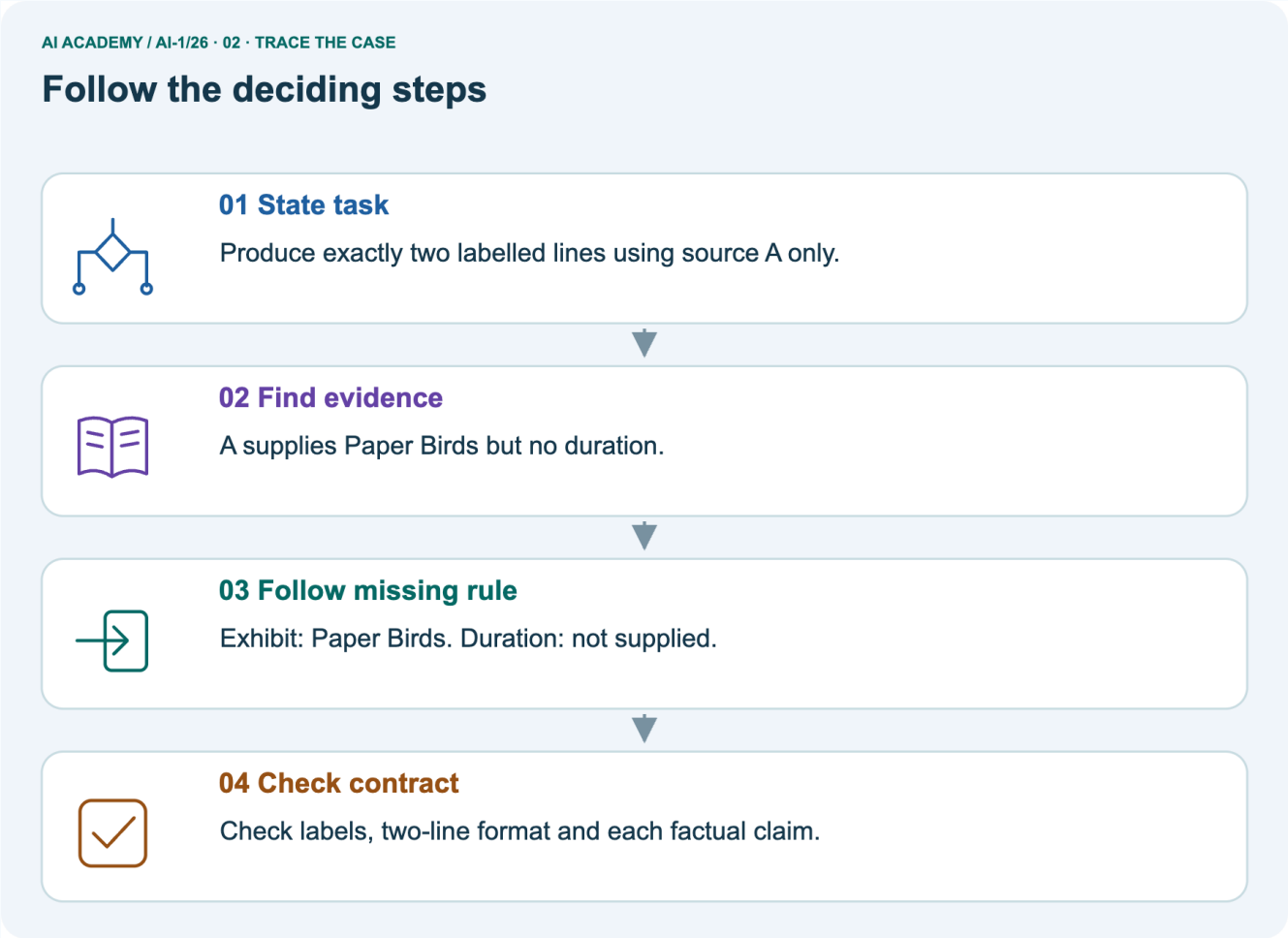
A fictional club needs a short exhibit card, without guesses. Supplied facts: Source A: exhibit Paper Birds, location Shelf 2. No duration appears.

Required **output**: two labelled lines, Exhibit and Duration; missing values must say not supplied.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

State task: Produce exactly two labelled lines using source A only. → Find **evidence**: A supplies Paper Birds but no duration. → Follow missing rule: Exhibit: Paper Birds.

Duration: not supplied. → Check contract: Check labels, two-line format and each factual claim.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. From prompt requirement to evidence of success

AI ACADEMY / AI-1/26 · 03 · INSPECT THE LESSON MODEL

From prompt requirement to evidence of success

Requirement	Expected behaviour	How to check
Use source C only	Monday and Room 7	Compare with C
Mark missing values	Time: not stated	Check absent time
Four labelled lines	Day Time Place Bring	Count and inspect
No added details	No invented promises	Check every claim

Choose these checks before reading a response. A response must satisfy all required criteria, not merely look tidy.

Follow the labelled picture with the matching explanation. Open the vector diagram

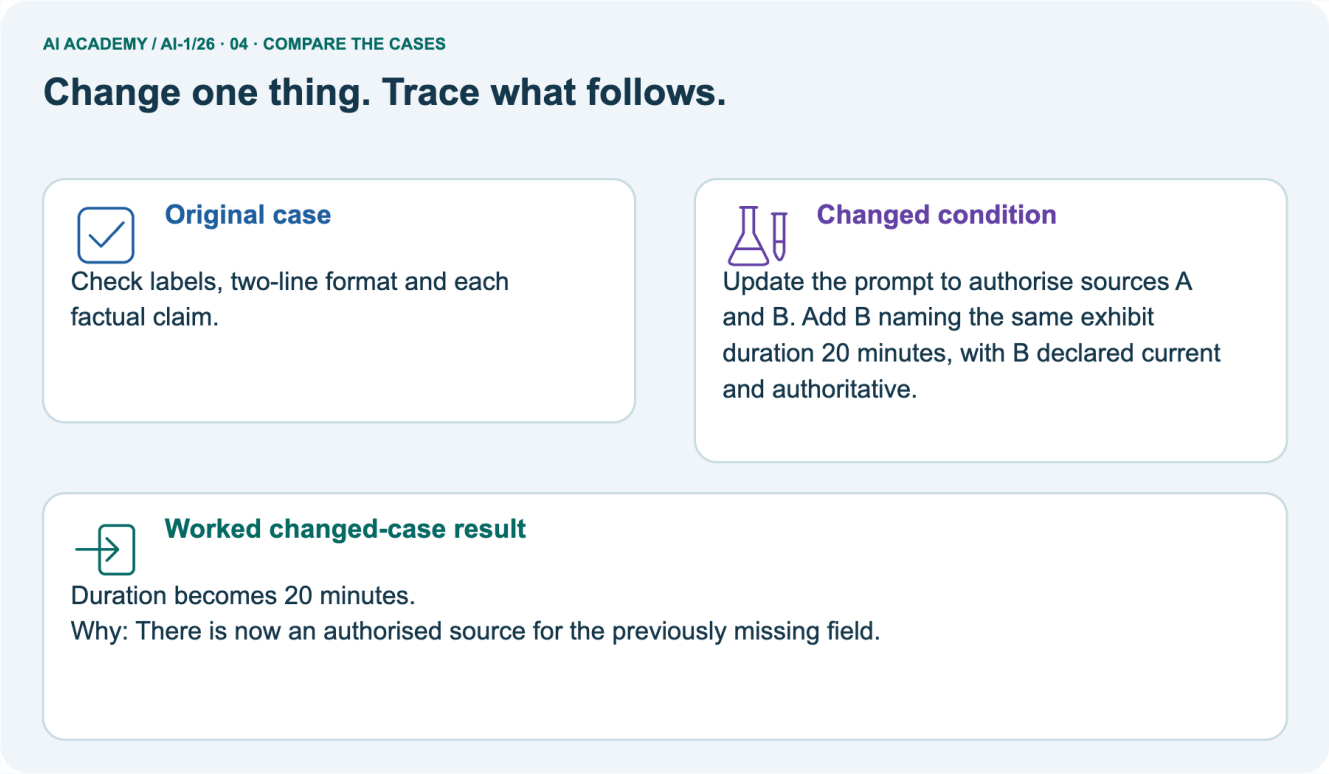
From **prompt** requirement to **evidence** of success. Use source C only | Monday and Room 7 | Compare with C. Mark missing values | Time: not stated | Check absent time.

Four labelled lines | Day Time Place Bring | Count and inspect. No added details | No invented promises | Check every claim

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

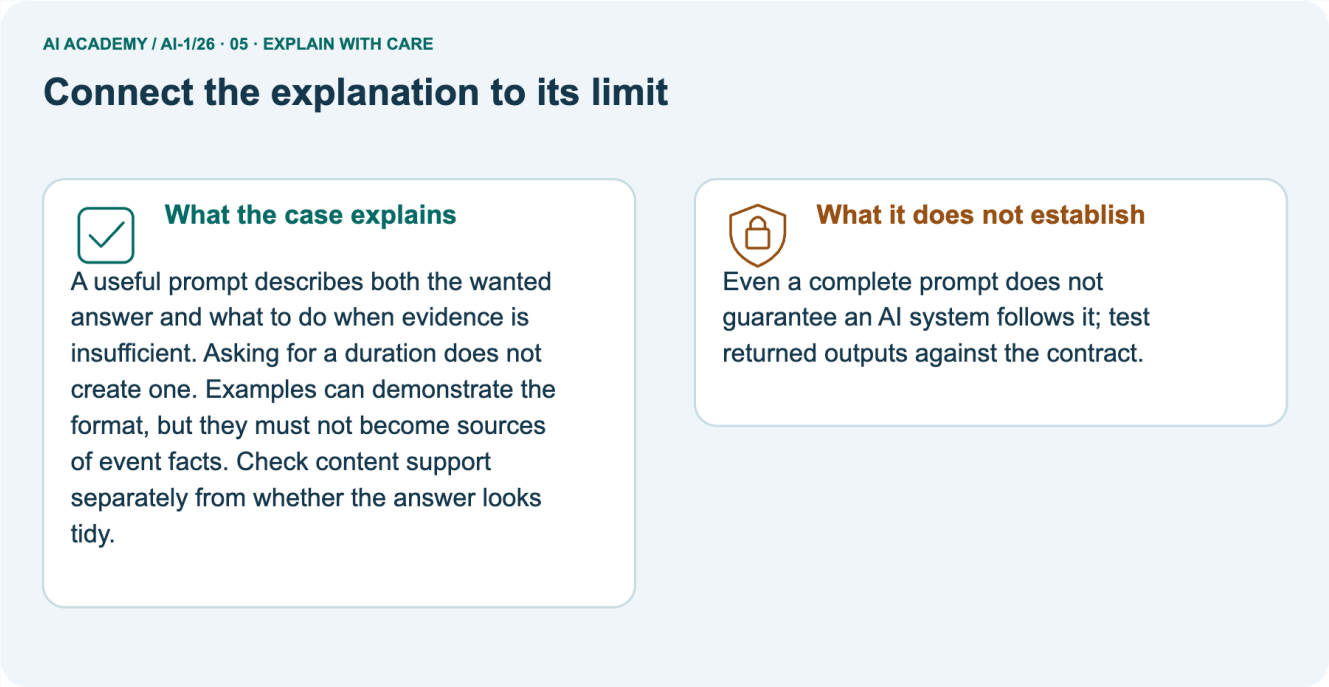
Original case: Check labels, two-line format and each factual claim. Changed condition: Update the prompt to authorise sources A and B. Add B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Expected result: Duration becomes 20 minutes. Reason: There is now an authorised source for the previously missing field.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

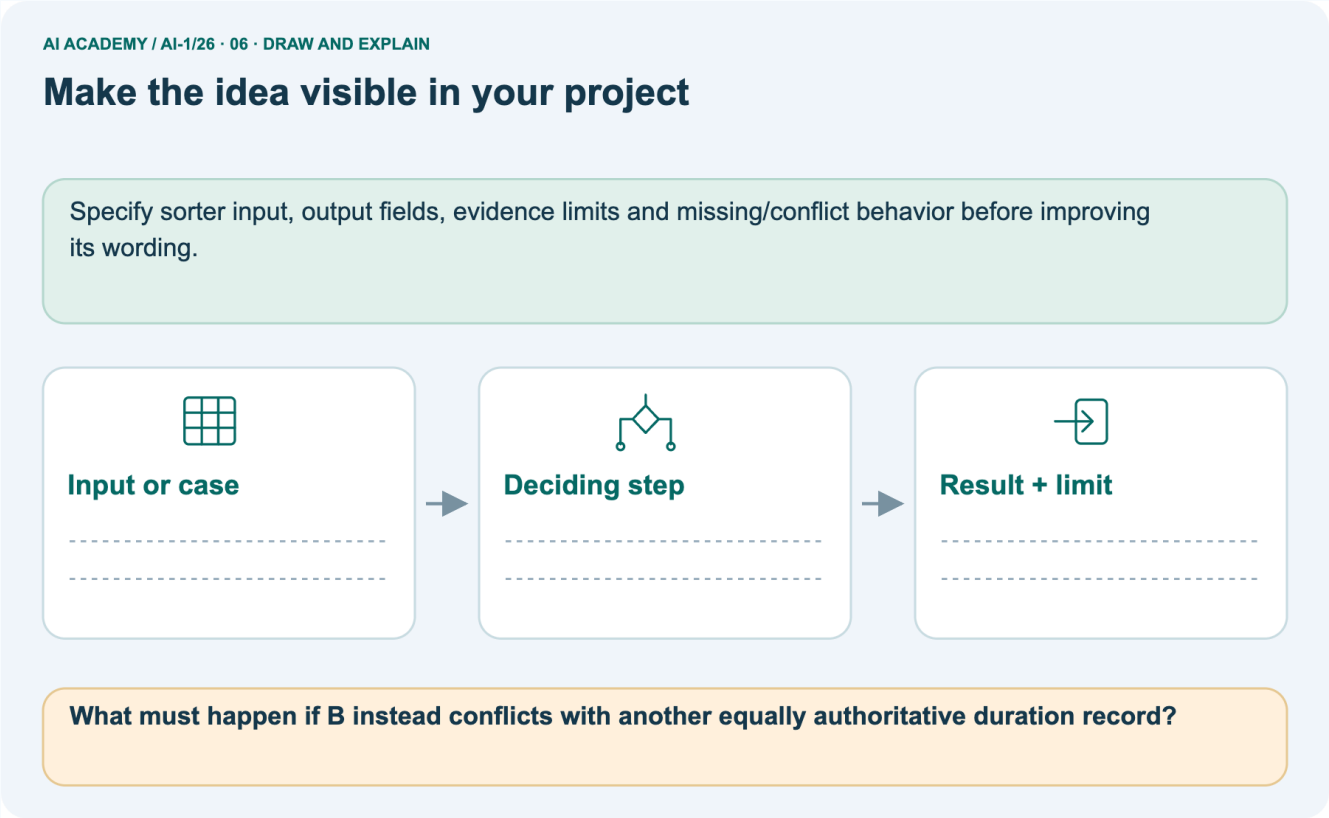
A useful **prompt** describes both the wanted answer and what to do when **evidence** is insufficient. Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts.

Check content support separately from whether the answer looks tidy. Limit: Even a complete **prompt** does not guarantee an AI system follows it; test returned **output**s against the contract.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Specify sorter **input**, **output** fields, **evidence** limits and missing/conflict behavior before improving its wording. Draw your **input**, deciding step, result and limit.

Supported question: What must happen if B instead conflicts with another equally authoritative duration record?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Prompts Are Design Instructions: the case

02. Follow the deciding steps

03. From prompt requirement to evidence of success

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.